

Dhiman Dasgupta

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Professional Summary

SDET with 2.5+ years of experience building scalable, CI/CD-driven automation frameworks for Web, Mobile (Android/iOS), and API platforms. Strong expertise in Playwright, TypeScript, Java, Selenide/Selenium, Appium, REST Assured, and GitHub Actions, with a focus on framework design, test stability, and scalable CI-driven execution. Proven track record of reducing regression cycles, increasing automation coverage, and delivering high-quality releases with near-zero production defects in fast-paced Agile product teams.

Core Competencies

- **Languages:** Java, TypeScript (Working), C/C++, SQL
- **Automation Frameworks:** Playwright, Selenium WebDriver, Selenide, TestNG, Maven, Apache POI
- **API Testing:** REST Assured, Postman, Playwright API Testing
- **Mobile Testing:** Appium (iOS & Android), TestMu AI (LambdaTest)
- **CI/CD & Tools:** Git, GitHub
- **AI Tools:** GitHub Copilot
- **Testing Practices:** Agile Testing, Regression Testing, Sanity Testing, Data-driven Testing, JIRA for defect tracking

Work Experience

Cognizant Technology Solutions

Bengaluru, India

Software Development Engineer in Test (SDET)

Sep 2023 – Present

- **Owned end-to-end quality engineering** for AI based Web, Mobile (Android/iOS), and **API platforms** in a client-facing product environment, ensuring stable monthly releases and high-quality user experience across platforms.
- Designed and maintained **unified automation frameworks (Web + Mobile + API)** using **Playwright, Selenide, Appium, and REST Assured**, reducing regression execution time by **34 hours per cycle** while improving overall test reliability.
- Increased automation coverage by **30%** by identifying high-impact user flows and edge cases using usage analytics and **AI-assisted scripting**, while reducing long-term maintenance through structured refactoring and clean code practices.
- Built and enhanced **CI/CD pipelines using GitHub Actions** with parallel execution, automated validation, and **Allure-based reporting**, while designing reusable utilities, request builders, response validators, and configuration layers using clean OOP principles.
- Implemented scalable mobile automation using **TestMu AI (LambdaTest) platform**, ensuring cross-device and **cross OS coverage for Android and iOS** applications.
- Proactively prevented production defects by combining **robust automated regression, exploratory testing, and effective defect triaging via JIRA**, maintaining a near-zero production defect rate across releases.

Key Projects

Flipkart End-to-End Test Automation Framework

[View on GitHub](#)

Java, Selenium, Selenide, TestNG, Maven, GitHub Actions, Allure

- Engineered a **production-grade UI automation framework** simulating real Flipkart user journeys.
- Implemented **CI/CD pipelines with headless execution, rich Allure reporting**, and automated failure analysis.
- Applied advanced synchronization, conditional validations, and POM architecture to eliminate flaky tests.
- Enabled **environment-based execution** and scalable test orchestration to simulate real CI production pipelines.

- Designed the framework to demonstrate **real-world SDET best practices** and CI maturity.

End-to-End API Automation Framework (DummyJSON)

[View on GitHub](#)

Java, Rest Assured, TestNG, GitHub Actions, Allure

- Architected a **scalable, maintainable API automation framework** to validate end-to-end backend workflows across Products, Users, Carts, and Orders, mirroring real production microservices.
- Integrated **CI/CD pipelines using GitHub Actions** with headless execution, failure triaging, and automated test reporting.
- Produced **diagnostic-rich Allure reports** with request/response payloads and assertion details to accelerate defect root-cause analysis.

Modern Automation Exposure (Playwright – Self-driven Implementation)

- Expanded automation expertise to modern frameworks by building proof-of-concept suites using **Playwright (TypeScript)** for high-performance UI and API testing.
- Implemented **auto-waiting and advanced locator strategies**, reducing flakiness and improving execution stability.
- Configured **cross-browser parallel execution (Chromium, Firefox, WebKit)** with optimized runtime performance.
- Performed backend validation using **Playwright Request Context** to enable combined UI + API testing workflows.
- Enabled **parallel test execution and headless CI integration** within GitHub Actions for faster feedback cycles.
- Designed a **lightweight, startup-oriented automation architecture** focused on speed, maintainability, and minimal boilerplate.
- Analyzed and optimized **Selenium vs Playwright execution models** to improve test speed, reliability, and developer experience.

Education

Bhilai Institute of Technology, Durg (CSVTU)

Jul 2023

B.Tech in Electronics & Telecommunication, CPI: 9.03

Delhi Public School, Bhilai (CBSE)

May 2019

Senior Secondary Education (Class 12th), Percentage: 81.4%

Certifications

Selenium Automation — API Automation — Mobile Automation - Udemy

Selenium Practitioner - Cognizant